Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
eakalabushkina@eakalabushkina:~$ ls /etc/ > file.txt
eakalabushkina@eakalabushkina:~$ ls >> file.txt
eakalabushkina@eakalabushkina:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

Рисунок 1: Запись в файл

Работа фильтра

```
^[[200~grep .conf file.txt > conf.txt
^[[201~eakalabushkina@eakalabushkgrep .conf file.txt > conf.txtonf.txt
eakalabushkina@eakalabushkina:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
makedumpfile.conf.sample
man_db.conf
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/eakalabushkina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
4/report/bib/cite.bib
/home/eakalabushkina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
5/report/_resources/csl
/home/eakalabushkina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
5/report/bib/cite.bib
/home/eakalabushkina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
6/report/_resources/csl
/home/eakalabushkina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
6/report/bib/cite.bib
/home/eakalabushkina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/_re
sources/csl
/home/eakalabushkina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/bib
/cite.bib
/home/eakalabushkina/git-extended/.git/hooks/commit-msg.sample
/home/eakalabushkina/git-extended/.git/objects/3f/c46da877d7d5b938c1c3a2cbe6f7409366daeb
/home/eakalabushkina/git-extended/.git/objects/be/cc1cff0cce5994994b4ad60d8cb289706fa637
/home/eakalabushkina/git-extended/.git/objects/cd
/home/eakalabushkina/git-extended/.git/objects/64/cfb3531e7d514b1ba3ddbf5d459d4ec1365c02
/home/eakalabushkina/git-extended/.git/config
/home/eakalabushkina/conf.txt
eakalabushkina@eakalabushkina:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/brltty/Text/he.ttb
/etc/brltty/Text/he_IL.ttb
/etc/brltty/Text/hi.ttb
/etc/brltty/Text/hr.ttb
/etc/brltty/Text/hu.ttb
/etc/brltty/Text/hy.ttb
/etc/containers/oci/hooks.d
/etc/foomatic/hashes.d
/etc/foomatic/hashes.d/hashes.new
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
/etc/logrotate.d/httpd
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
:find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
find: ‘/etc/userdb’: Permission denied
```

Рисунок 4: Поиск файлов

Управление процессами

```
eakalabushkina@eakalabushkina:~$
eakalabushkina@eakalabushkina:~$ find ~ -name "log*" > logfile &
[1] 6750
eakalabushkina@eakalabushkina:~$ rm logfile
[1]+  Done                    find ~ -name "log*" > logfile
eakalabushkina@eakalabushkina:~$
eakalabushkina@eakalabushkina:~$ kill 6750
bash: kill: (6750) - No such process
eakalabushkina@eakalabushkina:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
eakalabushkina@eakalabushkina:~$
eakalabushkina@eakalabushkina:~$ gedit &
[1] 6835
eakalabushkina@eakalabushkina:~$ ps | grep gedit
   6835 pts/1    00:00:00 gedit
eakalabushkina@eakalabushkina:~$ kill 6835
[1]+  Terminated              gedit
eakalabushkina@eakalabushkina:~$
eakalabushkina@eakalabushkina:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page  documents the GNU version of df.  df displays the amount of space available on the
       file system containing each file name argument.  If no file name is given, the space available on  all
       currently  mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environ-
       ment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If an argument is the absolute file name of a device node containing a mounted file system,  df  shows
       the  space  available  on  that file system rather than on the file system containing the device node.
       This version of df cannot show the space available on unmounted file systems, because on most kinds of
       systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all file systems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
              see SIZE format below
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes rather than device usage; although the apparent size is usually smaller,
              it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and
              the like

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
              see SIZE format below

       -b, --bytes
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
eakalabushkina@eakalabushkina:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91228044  63358292  60% /
devtmpfs         4011296        0   4011296   0% /dev
tmpfs            4035672       92   4035580   1% /dev/shm
tmpfs            1614272     1952   1612320   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035672       56   4035616   1% /tmp
/dev/nvme0n1p3 155186176 91228044  63358292  60% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop2         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807132      168    806964   1% /run/user/1059
/dev/sr0         2677920  2677920         0 100% /run/media/eakalabushkina/Fedora-WS-Live-43
eakalabushkina@eakalabushkina:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4       ./git-extended/.git/objects/98
4       ./git-extended/.git/objects/64
4       ./git-extended/.git/objects/ea
4       ./git-extended/.git/objects/6b
76      ./git-extended/.git/objects
8       ./git-extended/.git/logs/refs/heads
8       ./git-extended/.git/logs/refs/remotes/origin
8       ./git-extended/.git/logs/refs/remotes
16      ./git-extended/.git/logs/refs
20      ./git-extended/.git/logs
212     ./git-extended/.git
220     ./git-extended
0       ./monthly
0       ./reports/monthly/monthly
0       ./reports/monthly
0       ./reports
4       ./ski.plases/equipment
0       ./ski.plases/plans
4       ./ski.plases
0       ./australia
0       ./play/games/play
0       ./play/games
0       ./play
821800  .
eakalabushkina@eakalabushkina:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/eakalabushkina/git-extended/.git/objects/d9
/home/eakalabushkina/git-extended/.git/objects/3f
/home/eakalabushkina/git-extended/.git/objects/55
/home/eakalabushkina/git-extended/.git/objects/be
/home/eakalabushkina/git-extended/.git/objects/cd
/home/eakalabushkina/git-extended/.git/objects/56
/home/eakalabushkina/git-extended/.git/objects/98
/home/eakalabushkina/git-extended/.git/objects/64
/home/eakalabushkina/git-extended/.git/objects/ea
/home/eakalabushkina/git-extended/.git/objects/6b
/home/eakalabushkina/git-extended/.git/logs
/home/eakalabushkina/git-extended/.git/logs/refs
/home/eakalabushkina/git-extended/.git/logs/refs/heads
/home/eakalabushkina/git-extended/.git/logs/refs/remotes
/home/eakalabushkina/git-extended/.git/logs/refs/remotes/origin
/home/eakalabushkina/monthly
/home/eakalabushkina/reports
/home/eakalabushkina/reports/monthly
/home/eakalabushkina/reports/monthly/monthly
/home/eakalabushkina/ski.plases
/home/eakalabushkina/ski.plases/equipment
/home/eakalabushkina/ski.plases/plans
/home/eakalabushkina/australia
/home/eakalabushkina/play
/home/eakalabushkina/play/games
/home/eakalabushkina/play/games/play
eakalabushkina@eakalabushkina:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.